Paint Spraying Ladder – Easy-to-Read Version (FX5U kit, GX Works3)

Same program as the final ladder, written one rung at a time in plain layout. Type the rungs in this order. After each rung press **F4 (Convert)**. Uses only X0 to X7 and Y0 to Y7.

How to read a rung

Read left to right. All contacts in a row must be ON for the output on the right to act.

Symbol	Meaning	GX Works3 key
[X0]	Normally open contact: passes when X0 is ON	F5
[/ X0]	Normally closed contact: passes when X0 is OFF	F6
[LDP X0]	Rising edge: passes for one scan when X0 turns ON	type LDP X0
(M1)	Coil (output)	F7
[SET M7]	Switches M7 ON and keeps it ON	F8, type SET M7
[RST M7]	Switches M7 OFF	F8, type RST M7
<- OR	The row is in parallel (OR) with the row above	sF5 / sF6
[>= C0 D100]	Compare contact: passes when C0 is at least D100	type >= C0 D100

When a rung shows several output lines, enter each line as its own network, unless it says OR.

Devices you will meet

Real inputs: X0 entry sensor (button), X1 Stop (button), X2 Start (button), X3 exit sensor (button), X4 E-stop (toggle, ON = healthy), X5 door closed (toggle), X6 paint level OK (toggle), X7 pressure sensor live (toggle).

Real outputs (lamps): Y0 spray valve, Y1 conveyor, Y2 pump, Y3 fan, Y4 green, Y5 red, Y6 yellow, Y7 buzzer.

HMI bits (touch switches): M40 manual conveyor, M41 manual spray, M42 manual pump, M43 manual fan, M44 reject request, M47 reset totals, M48 maintenance acknowledge, M49 reset, M52 alarm acknowledge, M54 start, M55 stop, M100 jam, M101 manual mode, M102 auto mode, M103 pressure switch OK, M104 chamber clear (ON = empty).

Virtual outputs (HMI lamps): M70 door lock, M71 reject, M72 tower light.

Settings (HMI numbers): D100 batch size (15), D101 spray time in 0.1 s (250 = 25 s), D102 pressure setpoint (40 = 4.0 bar), D103 maintenance interval in batches (50), D104 purge time in 0.1 s (100 = 10 s). D110 = pressure from the pot. Leave D111 unused.

Timers (0.1 s units): T0 spray, T1 start delay K30, T2 purge, T3 door lock settle K30, T4 jam K50, T5 low pressure K50, T7 reject pulse K10, T8 discharge timeout K600.

Counters: C0 batch entry, C1 batch exit, C2 total painted, C3 completed batches, C4 rejected, C5 total incoming, C6 batches since service.

Suggested split for a team

Person	Rungs
1	Section A and B (R0 to R12)
2	Section C and D (R13 to R29)
3	Section E and F (R30 to R42)
4	Section G and H (R43 to R55)

The parts use the same device names, so they can be pasted into one project at the end.

SECTION A – System control

```
Rung 0: Kit inputs (4 separate networks)
|--[/ X1 ]-----------------------------( M61 )--|
|--[ X4 ]-----------------------------( M62 )--|
|--[ X3 ]-----------------------------( M63 )--|
|--[ SM400 ]-------------[ DIV SD6020 K40 D110 ]--|
  M61 = stop OK, M62 = E-stop OK, M63 = exit sensor.
  Last line turns the pot (0-10 V) into D110 = 0.0-10.0 bar.

Rung 1: Reset pulse
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|--[ M49 ]----------------------------[ PLS M50 ]--|

Rung 2: E-stop latch
|--[/ M62 ]---------------------------[ SET M7 ]--|

Rung 3: E-stop reset (E-stop released AND reset pressed)
|--[ M50 ]--[ M62 ]--------------------[ RST M7 ]--|

Rung 4: Auto mode
|--[ M102 ]--[/ M101 ]-----------------( M1 )--|

Rung 5: Manual mode
|--[ M101 ]--[/ M102 ]-----------------( M2 )--|

Rung 6: Ready to start
|--[ M62 ]--[/ M7 ]--[/ M6 ]--[ M1 ]-----( M9 )--|
|                             [ M2 ]   <- OR with M1

Rung 7: Start
|--[ X2 ]--[ M61 ]--[ M9 ]--------------[ SET M0 ]--|
|  [ M54 ]   <- OR with X2

Rung 8: Stop (any one condition switches the system off)
|--[/ M61 ]----------------------------[ RST M0 ]--|
|  [ M55 ]   <- OR
|  [ M6 ]    <- OR
|  [ M7 ]    <- OR
|  [/ M1 ]--[/ M2 ]   <- OR (both together)

Rung 9: Default settings (5 separate networks)
|--[ < D100 K1 ]----------[ MOV K15 D100 ]--|
|--[ < D101 K10 ]---------[ MOV K250 D101 ]--|
|--[ < D102 K1 ]----------[ MOV K40 D102 ]--|
|--[ < D103 K1 ]----------[ MOV K50 D103 ]--|
|--[ < D104 K1 ]----------[ MOV K100 D104 ]--|
  If a setting is empty or too small, its default is loaded.
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SECTION B – Object counting

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Rung 10: Count objects entering the batch (each X0 press = 1 object)
|--[ LDP X0 ]--[ Y1 ]--[ M0 ]--[ M1 ]--[/ M3 ]--[/ M11 ]--( C0 D100 )--|
  Counts only while the conveyor runs, and stops when the batch is full
  (M3) or while the batch is leaving (M11).

Rung 11: Total incoming objects
|--[ LDP X0 ]--[ Y1 ]--[ M0 ]------------( C5 K32767 )--|

Rung 12: Batch complete (15 objects counted)
|--[ >= C0 D100 ]--[/ M3 ]----------------[ SET M3 ]--|
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SECTION C – Conveyor

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Rung 13: Auto conveyor wanted + 3 s start delay (2 networks)
|--[ M0 ]--[ M1 ]--[/ M3 ]--[/ M4 ]--[/ M6 ]--[/ M7 ]--( M22 )--|
|                  [ M11 ]   <- OR with the M3 contact
|--[ M22 ]-----------------------------------------( T1 K30 )--|
  Conveyor is wanted while counting or leaving, never while a full
  batch waits for painting, and never while spraying.

Rung 14: Conveyor run request (auto after the delay, or manual)
|--[ M22 ]--[ T1 ]---------------------------------------( M5 )--|
|  [ M2 ]--[ M0 ]--[ M40 ]--[/ M4 ]--[/ M6 ]--[/ M7 ]   <- OR (whole row)

Rung 15: Conveyor motor (all safety conditions in series)
|--[ M5 ]--[/ M4 ]--[/ M6 ]--[/ M7 ]--[/ Y0 ]--[ M62 ]--( Y1 )--|

Rung 16: Jam timer
|--[ M100 ]--[ Y1 ]--------------------------------( T4 K50 )--|

Rung 17: Jam fault after 5 s
|--[ T4 ]-----------------------------------------[ SET M16 ]--|
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SECTION D – Painting

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Rung 18: Pressure OK
|--[ X7 ]--[ M103 ]--[ >= D110 D102 ]----------------( M21 )--|

Rung 19: Safe to paint
|--[ M3 ]--[ M0 ]--[ M1 ]--[ X5 ]--[ X6 ]--[ M62 ]--[/ M104 ]--[/ M6 ]--[/ M7
]--[/ Y1 ]--( M10 )--|
  Batch full, door closed, paint OK, objects in chamber (M104 OFF),
  no fault, conveyor stopped.

Rung 20: Start the painting sequence
|--[ M3 ]--[ M10 ]--[/ M8 ]--[/ M15 ]---------------[ SET M15 ]--|

Rung 21: Hazard hold (door locked and fan on until purge ends)
|--[ M15 ]-----------------------------------------( M13 )--|
|  [ M8 ]--[/ M11 ]   <- OR

Rung 22: Exhaust fan
|--[ M13 ]-----------------------------------------( Y3 )--|
|  [ M2 ]--[ M0 ]--[ M43 ]   <- OR (manual fan)

Rung 23: Door lock + 3 s settle timer (2 networks)
|--[ M13 ]--[ X5 ]---------------------------------( M70 )--|
|  [ M30 ]--[ X5 ]   <- OR
|--[ M70 ]-----------------------------------------( T3 K30 )--|

Rung 24: Paint pump
|--[ M15 ]--[ M70 ]--[ Y3 ]--[ M0 ]--[/ M6 ]--[/ M7 ]--[ M62 ]--( Y2 )--|
|  [ M2 ]--[ M0 ]--[ M42 ]--[ X6 ]--[/ M6 ]--[/ M7 ]--[ M62 ]   <- OR (manual
pump)

Rung 25: Automatic spray + 25 s timer (3 networks)
25a: start spraying
|--[ M15 ]--[ M0 ]--[/ M8 ]--[/ M6 ]--[/ M7 ]--[ T3 ]--[ Y2 ]--[ M21 ]--[ SET
M24 ]--|
25b: stop spraying (any one condition)
|--[ M8 ]------------------------------------------[ RST M24 ]--|
|  [ M6 ]    <- OR
|  [ M7 ]    <- OR
|  [/ M0 ]   <- OR
|  [/ M15 ]  <- OR
25c: spray timer (25 s = D101 x 0.1 s)
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|--[ M24 ]----------------------------------------( T0 D101 )--|

Rung 26: Manual spray command
|--[ M2 ]--[ M0 ]--[ M41 ]--[ X5 ]--[ X6 ]--[/ M6 ]--[/ M7 ]--[/ Y1 ]--[/ M15
]--( M30 )--|

Rung 27: Spraying active (auto or manual)
|--[ M24 ]----------------------------------------( M4 )--|
|  [ M30 ]   <- OR

Rung 28: Spray valve (all safety conditions in series)
|--[ M4 ]--[ M70 ]--[ T3 ]--[ Y2 ]--[ M21 ]--[ X5 ]--[ M62 ]--[/ M6 ]--[/ M7
]--[/ Y1 ]--( Y0 )--|

Rung 29: Spray complete (2 networks)
|--[ T0 ]-----------------------------------------[ SET M8 ]--|
|--[ T0 ]-----------------------------------------[ RST M15 ]--|
  Valve and pump stop. The fan keeps running for the purge.
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SECTION E – Batch discharge

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Rung 30: Purge timer (10 s = D104 x 0.1 s)
|--[ M8 ]--[/ M11 ]---------------------------------( T2 D104 )--|

Rung 31: Start discharge (fan off, door unlocks, conveyor restarts)
|--[ T2 ]-----------------------------------------[ SET M11 ]--|

Rung 32: Count objects leaving (each X3 press = 1 object)
|--[ LDP M63 ]--[ M11 ]-----------------------------( C1 D100 )--|

Rung 33: Batch fully discharged
|--[ >= C1 D100 ]--[ M104 ]--[ M11 ]-----------------( M14 )--|

Rung 34: Reset for the next batch (5 networks, each starts with M14)
|--[ M14 ]-----------------------------------------[ RST C0 ]--|
|--[ M14 ]-----------------------------------------[ RST C1 ]--|
|--[ M14 ]-----------------------------------------[ RST M3 ]--|
|--[ M14 ]-----------------------------------------[ RST M8 ]--|
|--[ M14 ]-----------------------------------------[ RST M11 ]--|
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SECTION F – Production monitoring

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Rung 35: Total painted (good) objects
|--[ LDP M63 ]--[ M11 ]--[/ M46 ]--------------------( C2 K32767 )--|

Rung 36: Completed batches
|--[ LDP M14 ]----------------------------------------( C3 K32767 )--|

Rung 37: Reject an object (2 networks, type the counter FIRST)
|--[ LDP M44 ]--[ M11 ]-------------------------------( C4 K32767 )--|
|--[ LDP M44 ]--[ M11 ]--[/ M46 ]---------------------[ SET M46 ]--|

Rung 38: Reject pulse, 1 s (2 networks)
|--[ M46 ]--------------------------------------------( T7 K10 )--|
|--[ M46 ]--------------------------------------------( M71 )--|

Rung 39: End of reject pulse
|--[ T7 ]---------------------------------------------[ RST M46 ]--|

Rung 40: Maintenance counter (batches since service)
|--[ LDP M14 ]----------------------------------------( C6 D103 )--|

Rung 41: Maintenance reminder
|--[ C6 ]---------------------------------------------( M20 )--|

Rung 42: Reset totals / acknowledge maintenance (5 networks)
|--[ M47 ]--[/ M0 ]-----------------------------------[ RST C2 ]--|
|--[ M47 ]--[/ M0 ]-----------------------------------[ RST C3 ]--|
|--[ M47 ]--[/ M0 ]-----------------------------------[ RST C4 ]--|
|--[ M47 ]--[/ M0 ]-----------------------------------[ RST C5 ]--|
|--[ M48 ]--------------------------------------------[ RST C6 ]--|
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SECTION G – Faults and alarms

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Rung 43: Low paint fault
|--[ M0 ]--[/ X6 ]------------------------------------[ SET M17 ]--|

Rung 44: Low pressure timer (pump on, pressure not OK)
|--[ Y2 ]--[/ M21 ]-----------------------------------( T5 K50 )--|

Rung 45: Low pressure fault after 5 s
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|--[ T5 ]-----------------------------------------------[ SET M18 ]--|

Rung 46: Discharge timeout timer (60 s)
|--[ M11 ]--[/ M51 ]-------------------------------------( T8 K600 )--|

Rung 47: Abnormal process fault (timeout, or spray and conveyor both ON)
|--[ T8 ]-----------------------------------------------[ SET M51 ]--|
|  [ Y0 ]--[ Y1 ]   <- OR

Rung 48: Clear the latched faults (5 networks, each needs Reset M50 plus the
cause fixed)
|--[ M50 ]--[/ M100 ]-----------------------------------[ RST M16 ]--|
|--[ M50 ]--[ X6 ]--------------------------------------[ RST M17 ]--|
|--[ M50 ]--[/ T5 ]--[ M21 ]----------------------------[ RST M18 ]--|
|--[ M50 ]--[ X5 ]--------------------------------------[ RST M19 ]--|
|--[ M50 ]--[/ T8 ]--------------------------------------[ RST M51 ]--|

Rung 49: Door open fault
|--[ M0 ]--[/ X5 ]--------------------------------------[ SET M19 ]--|

Rung 50: General fault (any one fault)
|--[ M16 ]-----------------------------------------------( M6 )--|
|  [ M17 ]   <- OR
|  [ M18 ]   <- OR
|  [ M19 ]   <- OR
|  [ M51 ]   <- OR

Rung 51: Silence the buzzer (3 networks)
|--[ M52 ]--[ M6 ]--------------------------------------[ SET M53 ]--|
|--[ M52 ]--[ M7 ]--------------------------------------[ SET M53 ]--|
|--[/ M6 ]--[/ M7 ]-------------------------------------[ RST M53 ]--|

Rung 52: Buzzer (steady for E-stop, beeping for faults)
|--[ M7 ]--[/ M53 ]-------------------------------------( Y7 )--|
|  [ M6 ]--[ SM412 ]--[/ M53 ]   <- OR
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SECTION H – Indicators

```
Rung 53: Green and red lamps (2 networks)
|--[ M0 ]--[/ M6 ]--[/ M7 ]----------------------------( Y4 )--|
|--[ M6 ]-----------------------------------------------( Y5 )--|
|  [ M7 ]   <- OR

Rung 54: Yellow lamp (painting or discharging)
|--[ M13 ]----------------------------------------------( Y6 )--|
|  [ M11 ]   <- OR

Rung 55: Tower light (steady = normal, blinking = fault or maintenance)
|--[ Y4 ]--[/ M20 ]-------------------------------------( M72 )--|
|  [ M6 ]--[ SM412 ]    <- OR
|  [ M7 ]--[ SM412 ]    <- OR
|  [ M20 ]--[ SM412 ]   <- OR

|-----------------------------------------------------[ END ]--|
```

Notes

- **Changed from the earlier version:** R25 uses SET and RST instead of one coil, which is easier to read and behaves the same. R37 asks for the counter first, because SET M46 would block the count if it came first.
- **Order matters:** keep the rungs in this order, especially R25 to R31.
- **Check SD6020** (analog input CH1) in the FX5U manual before relying on the pot.
- **E-stop at power-up:** X4 must be ON, otherwise E-stop is latched. Turn X4 ON and press Reset on the HMI.
- **Safety:** real E-stop and door-lock release must also be hardwired through a safety relay. The PLC only monitors them.

Quick test on the kit

1. Toggles X4, X5, X6, X7 ON. HMI: M102 ON, M101 OFF, M103 ON, M104 ON. Turn the pot above 4 V.
2. Press X2. Green Y4 comes on. Y1 starts after 3 s.
3. Press X0 fifteen times. Y1 stops.

4. Set M104 OFF. Fan Y3 and pump Y2 start. After 3 s the spray Y0 and yellow Y6 run for 25 s.
5. The spray and pump stop. The fan runs 10 s more, then Y1 restarts.
6. Press X3 fifteen times, then set M104 ON. C3 becomes 1.